

PRE-ASSIGNMENT-3

The dataset is divided into training, development, and test sets while maintaining speaker diversity in terms of age, gender distribution, and PHQ-8 scores. The training and development sets include a mix of Wizard-of-Oz (WoZ) and AI-controlled interactions, whereas the test set consists solely of AI-controlled sessions. Sessions with IDs [300,492] were conducted with a WoZ-controlled agent, while sessions with IDs [600,718] were collected using an AI-controlled agent.

The train/dev/test splits are available in the labels directory. Each split file contains:

- **Participant_ID**
- **Gender**
- **PHQ_Binary**
- **PHQ_Score**
- **PCL-C (PTSD Score)**
- **PTSD Severity**

Additionally, *Detailed_PHQ8_Labels.csv* provides responses to individual PHQ-8 questions, allowing for symptom-specific analysis. For instance, the dataset includes ratings for symptoms like sleep difficulties.

Responses to specific questions can be found by navigating the tabs on the left.

Name: POOJAL KATIYAR(220770).

Ma'am it might be possible that my final Assignment could be delayed due to health conditions about which I have already mailed.

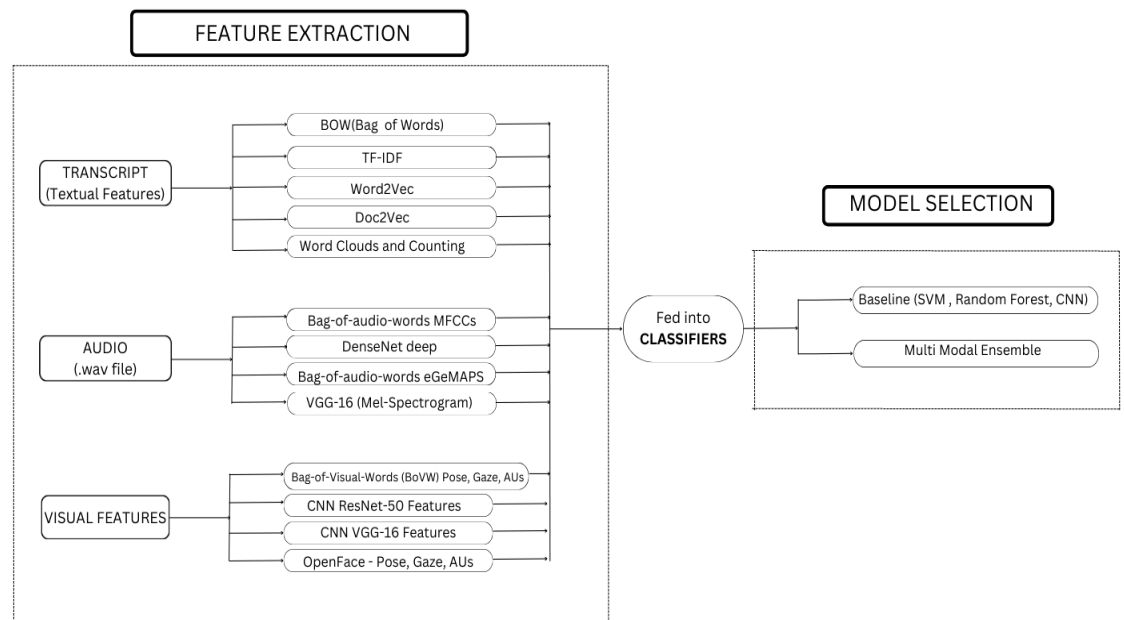
1) What are your specific aims towards the overarching goal?

- Develop a real-time prediction model to assess stress or negative mood from multimodal data (text, audio, visual).
- Improve Stress/Negative Mood Detection Accuracy – Develop a robust ensemble model that enhances the accuracy of detecting stress or negative mood in real-time.
- Utilize Multimodal Data – Leverage textual, audio, and visual features to improve model performance and identify key stress indicators.
- Balance Dataset Bias – Implement techniques like undersampling and repeated k-cross validation to mitigate class imbalance issues.

- Enhance Interpretability – Provide insights into linguistic patterns, facial expressions, and vocal cues associated with stress or negative mood.
- Optimize Generalization – Use ensemble learning to reduce overfitting and ensure better real-world applicability.
- Will be doing supervised ML like SVM, Logistic or using ensemble learning on training dataset. Then will see which of the models give a better F1 score.

2) What kind of features will you be using?

This is a brief flow chart that I prepared. I will try to use transcript, .wav audio file and multimodal features which are provided in the dataset. I will extract relevant features. I plan to follow the below mentioned flowchart where some features are already extracted for each participants. Can Perform Principal Component Analysis (PCA) / Feature Selection for Dimensionality Reduction



I will use the following features to predict stress or negative mood:

Text Features:

- Transcripts from interviews
- Sentiment analysis (positive/negative tone)
- Linguistic features (word frequency, complexity, sentiment scores)
- Topic modeling (identifying stress-related topics)

Audio Features:

- Prosodic features (pitch, intensity, speech rate)
- Voice quality (jitter, shimmer, HNR)
- Emotion-related features (extracted using eGeMAPS, MFCCs,)
- VGG-16

Visual Features:

- Facial expressions (OpenFace extracted features)
- Gaze direction, blink rate
- Head movements and gestures
- CNN ResNet and VGG

Graph-based Features – Derived from speaker turns and conversational flow to analyze interaction patterns.

3) And by using what kind of methods do you aim to achieve your aims?

Dataset Challenges

Small Sample Size

- The dataset contains 275 participants, making it prone to overfitting in machine learning models.

Class Imbalance

- There are more than twice the number of non-depressed individuals compared to depressed individuals.

- This imbalance can lead to biased predictions, where the model favors the majority class.

Choice of Supervised Learning

- Supervised learning is the best approach since labels (PHQ-8 scores and PCL-C scores) are provided.

Modeling Approaches:

- Single Modality: Train separate SVM, LSTM, CNN, or Random Forest models.
- Multimodal Fusion:
 - Early Fusion: Concatenate features before passing to a deep model (e.g., Transformer).
 - Late Fusion: Train separate models and combine predictions via ensemble learning (majority voting, weighted averaging).

Evaluation: k-Fold cross-validation, F1-score for classification, RMSE for regression.

Why K-Cross Validation?

- The primary goal is to detect true positive cases (individuals with stress).
- Recall (Sensitivity) is crucial, but predicting all cases as stressed would lead to:
 - High recall (detects all stressed individuals).
 - High false positives (misclassified non-stressed cases).
- Solution:
 - Use K-Cross Validation to ensure generalization on unseen data.
 - Select the model with the highest F1-score to balance precision & recall.

Undersampling for Class Balancing

- To reduce class imbalance, create a 50:50 ratio of depressed vs. nondepressed cases and for maybe PTSD also and compare which one is better.

Hyperparameter Tuning

- Fine-tune hyperparameters for each classifier using k-cross validation to improve performance.

Ensemble Model Approach

Why Use an Ensemble?

- Combines predictions from multiple classifiers to improve robustness.
- Reduces bias from individual classifiers.
- Increases stability & generalization of the model.

Final Prediction Strategy

- Final prediction = Average probability output of selected classifiers.
- Choose the ensemble with the highest F1-score.

This is my current expected way to work with the DAIC dataset which may be affected when working with the data.

4.Resources(These are what I found relevant with respect to the multimodal Ensemble that I plan to do. This list is not exhaustive. After I finish with the assignment whatever more resources I will use, I will write them in the final submission itself):

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC9689708/>
- <https://www.nature.com/articles/s44184-024-00112-8>
- <https://www.proquest.com/docview/3100732557?fromopenview=true&pq-origsite=gscholar&sourcetype=Dissertations%20&%20Theses>
- <https://ruor.uottawa.ca/items/d1a3942e-70e9-45ba-baed-82649d1e557d>
- <https://www.sciencedirect.com/science/article/pii/S1532046423000205>
- https://www.researchgate.net/publication/389915421_Mental_Stress_Detection_with_the_Multimodal_Data_Using_Ensemble_Optimization_Enabled_Explained_Convolutional_Neural_Network
- <https://link.springer.com/article/10.1007/s00500-025-10441-1>
- <https://link.springer.com/article/10.1007/s44174-025-00296-3>

I have written quite a lot of things which I plan to do. But it might be possible due to my current health conditions that I may not be able to keep up with them, but I will try my best to complete this assignment with the best possible efforts.